

Yutian Tang

*Assistant Professor
(UK Lecturer)
Senior Member, IEEE*

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Research Interests

Software Engineering (AIGC for SE, AI4SE, SE4AI, Program Analysis, Empirical Software Engineering, Software Component Analysis, Android);
Security (Mobile Security, Fuzzing, Software Supply Chain Security)

Profile

Dr. Yutian Tang is an Assistant Professor (UK Lecturer) at the School of Computing Science, University of Glasgow, United Kingdom (PI, Ph.D. Supervisor). He received his Ph.D. degree from the Department of Computing, The Hong Kong Polytechnic University. His research interests include AI+SE (Large-language Model+SE), program analysis, empirical software engineering, privacy protection, and software testing. He is a senior member of IEEE, a member of ACM, CCF, European Alliance for Innovation (EAI), and EuroSys. He is also a member of EPSRC Peer Review College, a technical committee member of the IEEE Technical Committee on Software Engineering. He is also affiliated with Trustworthy Digital Identity Interest Group and Interpretation, Verification and Modelling Interest Group, The Alan Turing Institute. He also contributes to the IEEE Standards Association P1633 Working Group (IEEE Recommended Practice on Software Reliability).

Education

- 2023-2025 **Postgraduate Certificate in Academic Practice**, University of Glasgow, UK
2013-2018 **Ph.D.**, Department of Computing, The Hong Kong Polytechnic University, Hong Kong SAR, China
Advisor ○ Full Prof. Xiapu Luo, and Dr. Hareton Leung (Associate Professor Emeritus)
2009-2013 **B.Sc.**, Department of Computer Science and Technology, Jilin University, China

Research Positions

- 2023-Present **Assistant Professor (UK Lecturer)**,
*School of Computing Science,
University of Glasgow,
Leading University in UK, Member of Russel Group, Top-100 University in the World
Glasgow, Scotland, United Kingdom*

- 2020-2023 **Assistant Professor**, *School of Information Science and Technology, ShanghaiTech University, China*
- 2018-2020 **Post-doc Research Fellow**, *The Hong Kong Polytechnic University, Hong Kong SAR, China*

Visiting Positions

- 2025-2025 **Visiting Professor**,
*Department of Computer Science, Host: Prof. Per Runeson
Lund University
Lund, Sweden (Under Get Started Funding Scheme)*

Grants

Highly competitive grants are highlighted

Summary I have secured 8 research grants as solo PI or leading PI, which collectively amount to 418,836 (GBP)/553,405 (USD) in total. This sum does not include any start-up grants.

As Solo or Leading Principal Investigator (8)

- LG8 OpenAI Cybersecurity Grant
Agency/Company: OpenAI
Role: PI,
Contract period: 2025.09-2026.09
Share: 100%
- LG7 Get Started Funding
Agency/Company: Lund University
Role: PI, with Professor Per Runeson from Lund University
Contract period: 2025.09-2026.09
Share: 100%
- LG6 Google Cloud for Researchers Program
Agency/Company: Google
Role: Solo PI
Contract period: 2024.09-2025.09
Share: 100%
- LG5 OpenAI's Researcher Access Program
Agency/Company: OpenAI
Role: Solo PI
Contract period: 2024.03-2025.03
Share: 100%
- LG4 Start-up Research Grant, University of Glasgow
Agency/Company: University of Glasgow
Role: PI
Collaborators: Yutian Tang (PI)
Contract period: 2023-2025
Share: 100%

LG3 Research on Android App Debloating with Multi-dimension, Multi-semantic Module Mining (No. 6220072429)
Agency/Company: National Natural Science Foundation of China (NSFC) -Youth Program
Role: PI
Collaborators: Yutian Tang (PI)
Contract period: 2023.01-2025.12
Share: 100%
Success Rate: 17.29%

LG2 Vulnerable Components Detection for Apps (No. 21PJ1410700)
Agency/Company: Shanghai Pujiang Program, Science and Technology Commission of Shanghai Municipality
Role: PI
Collaborators: Yutian Tang (PI)
Contract period: 2021.10-2023.09
Share: 100%

LG1 Early Career Research Fund (No.2020F0203-000-14)
Agency/Company: ShanghaiTech University
Role: PI
Collaborators: Yutian Tang (PI)
Contract period: 2020-2023
Share: 100%

As Participants (2)

PG2 Improving Mobile Users' Privacy Decisions via Exposing Unexpected Sensitive Operations in Hardened Apps
Agency/Company: Research Grants Council GRF, Hong Kong
Role: Participant
Collaborators: Xiapu Luo (PI)
Contract period: 2018-2020
Share: Not available

PG1 Research and Development of An Advanced IPS System for Aftermarket Telematics
Agency/Company: Innovation and Technology Fund (ITF), Hong Kong
Role: Participant
Collaborators: Xiapu Luo (PI)
Contract period: 2018-2020
Share: Not available

Publications

Summary According to Google Scholar, my works attracted more than 1,441 citations, my h-index is 18. In the domain of software engineering, the most prestigious venues are IEEE TSE, ACM TOSEM, ASE, OOPSLA, ICSE, ISSTA, ESEC/FSE, CCS, and WWW.

Google Scholar <https://scholar.google.com/citations?user=cgdLmTUAAAAJ>

Top-tier Conf./Journal (21) ESEC/FSE (1); ICSE (1); WWW (1); ASE (4); IEEE TSE (10), ACM TOSEM (1), OOPSLA (2), ISSTA (1);

JCR-Q1 (16), Core-A/A*(18), CCF-A (21), CORE-B(9), CORE-C(1), CCF-B(16), CCF-C(6)

★ for the corresponding author, (stu) for my students under my supervision

Selected Papers are highlighted

Pre-print Papers (1)

- P.2 Zhichao Zhou, **Yutian Tang**, Yun Lin, and Jingzhu He. "An LLM-based Readability Measurement for Unit Tests' Context-aware Inputs." *arXiv preprint arXiv:2407.21369* (2024).

Refereed Conference Papers (26);

- C.26** Zeyu Lu, Peng Zhang, Yuge Nie, Yibiao Yang, **Yutian Tang**, Chun Yong Chong, Yuming Zhou, "Beyond Coverage: Automatic Test Suite Augmentation for Enhanced Effectiveness Using Large Language Models ", *International Conference on Object-Oriented Programming Systems, Languages, and Applications (OOPSLA)*,2026 **CORE-A*,CCF-A**;
- C.25** Zhaoyu Qiu, Ming Fan, Bocan Ma, **Yutian Tang**, Lei Xue, Haijun Wang, Ting Liu, "WhisperCatcher: Demystifying Unauthorized and Encrypted Private Data Transmission in Android Applications", *IEEE/ACM International Conference on Software Engineering (ICSE)*,2026 **CORE-A*,CCF-A,ACM SIGSOFT Distinguished Paper Award**;
- C.24 David Romero OrganvÍdez, Oscar Díaz, **Yutian Tang**, David Benavides. "Feedback Analysis in Software Product Line Forked Developments", *ACM International System and Software Product Line Conference (SPLC)*,2025;
- C.23** Zhijie Liu^(stu.), Qiyi Tang , Sen Nie, Shi Wu, Liangfeng Zhang, **Yutian Tang***. "KEENHash: Hashing Programs into Function-aware Embeddings for Large-scale Binary Code Similarity Analysis", *ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA)*,2025 **CORE-A,CCF-A**;
- C.22 **Yutian Tang**, Hongchen Cao, Yuxi Chen, David Lo. "Characterising Bugs in Jupyter Platform", *International Conference on Evaluation and Assessment in Software Engineering (EASE)*,2025 **CORE-A,CCF-B**;
- C.21** Yuchen Ji, Ting Dai, Zhichao Zhou, **Yutian Tang**,Jingzhu He. "Artemis: Toward Accurate Detection of Server-Side Request Forgeries through LLM-Assisted Inter-Procedural Path-Sensitive Taint Analysis", *The Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA)*,2025 **CORE-A, CCF-A, Top 1 conference in SE**;
- C.20 Yuchen Ji, Ting Dai, **Yutian Tang**, Jingzhu He, "Whether We Are Good Enough to Detect Server-Side Request Forgeries in PHP-native Applications?", *ACM Conference on Computer and Communications Security (CCS) Poster Track*,2024;

- C.19** Zhijie Liu^(stu.), Liangfeng Zhang, **Yutian Tang***, “Enhancing Malware Detection for Android Apps: Detecting Fine-granularity Malicious Components”, *IEEE/ACM Automated Software Engineering (ASE) Conference, 2023*: pp. 1212–1224 **CORE-A***, **CCF-A**, **Top 3 conference in SE**;
- C.18** Zhichao Zhou^(stu.), Yuming Zhou, Chunrong Fang, Zhenyu Chen, **Yutian Tang***, “Selectively Combining Multiple Coverage Goals in Search-Based Unit Test Generation”, *35th IEEE/ACM International Conference on Automated Software Engineering (ASE), 2022*: pp. 1-12 **CORE-A***, **CCF-A**, **Top 3 conference in SE**;
- C.17** Guosheng Xu, Siyi Li, Hao Zhou, Shucen Liu, **Yutian Tang**, Li Li, Xiapu Luo, Xusheng Xiao, Guoai Xu and Haoyu Wang, “Lie to Me: Abusing the Mobile Content Sharing Service for Fun and Profit”, *The Web Conference (WWW), 2022*: pp. 3327–3335, **CORE-A***, **CCF-A**;
- C.16 Zejun Wu, Chao Ma, Xiaochuan Shi, Libing Wu, Dian Zhang, **Yutian Tang**, “BRNN-GAN: Generative Adversarial Networks with Bi-directional Recurrent Neural Networks for Multivariate Time Series Imputation”, *IEEE International Conference on Parallel and Distributed Systems, 2021*: pp. 217-224, **CORE-B**, **CCF-C**;
- C.15 Kunsong Zhao, Zhou Xu, Meng Yan, **Yutian Tang**, Ming Fan, Gemma Catolino, “Just-in-Time Defect Prediction for Android Apps via Imbalanced Deep Learning Model”, *The 36th ACM/SIGAPP Symposium On Applied Computing, 2021*: pp 1447-1454.
- C.14** Hao Zhou, Haoyu Wang, Yajin Zhou, Xiapu Luo, **Yutian Tang**, Lei Xu, Ting Wang, “Demystifying Diehard Android Apps”, *35th IEEE/ACM International Conference on Automated Software Engineering (ASE), 2020*: pp. 187-198 **CORE-A***, **Top 3 conference in SE**, **CCF-A**;
- C.13 Kunsong Zhao, Zhou Xu, Tao Zhang, **Yutian Tang**, “Simplified Deep Forest Model based Just-In-Time Defect Prediction for Android Mobile Apps”, *20th International Conference on Software Quality, Reliability, and Security (QRS), 2020*, **CORE-B**, **CCF-C**;
- C.12** **Yutian Tang**, Yulei Sui, Haoyu Wang, Xiapu Luo, Hao Zhou, Zhou Xu, “All Your App Links (are) Belong to Us: Understanding the Threats of Instant Apps based Attacks”, *The ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), 2020*, **CORE-A***, **Top 2 conference in SE**, **the vulnerability reported is confirmed by Google**, **CCF-A**;
- C.11** Yan Cai, **Yutian Tang**, Haicheng Li, Le Yu, Hao Zhou, Xiapu Luo, Liang He, Purui Su, “Resource Race Attacks on Android”, *27th IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), 2020*, , **CORE-A**, **the vulnerability reported is confirmed by Google**, **CCF-B**;
- C.10** **Yutian Tang**, Xian Zhan, Hao Zhou, Xiapu Luo, Zhou Xu, Yajin Zhou, Qiben Yan, “Demystifying Application Performance Management Libraries for Android”, *The 34th IEEE/ACM International Conference on Automated Software Engineering (ASE), pp. 682-685, 2019*, **CORE-A***, **CCF-A**;

- C.9 Zhou Xu, Sizhe Ye, Tao Zhang, Zhen Xia, Shuai Pang, Yong Wang, **Yutian Tang**, "MVSE: Effort- Aware Heterogeneous Defect Prediction via Multiple-View Spectral Embedding", 19th International Conference on Software Quality, Reliability, and Security (QRS), pp. 10-17, 2019, CORE-B,CCF-C
- C.8 Zhou Xu, Shuai Pang, Yifeng Zhang, **Yutian Tang**, Jin Liu, Xiapu Luo, Jacky Keung, Xiaohui Cui, "Identifying Crashing Fault Residence Based on Cross Project Model", International Symposium on Software Reliability Engineering (ISSRE), pp. 183-194, 2019, **CORE-A,CCF-B**;
- C.7 Xian Zhan, Tao Zhang, **Yutian Tang**, "A Comparative Study of Android Repackaged Apps Detection Techniques", 2019 IEEE International Conference on Software Analysis, Evolution, and Reengineering (SANER), pp. 321-331, 2019, **CORE-A,CCF-B**;
- C.6** Xuejiao Zhao, Hongwei Li, **Yutian Tang**, Dongjing Gao, Lingfeng Bao, Chinghung Lee, "A Smart Context-aware Program Assistant based on Dynamic Programming Event Modeling", The 29th IEEE International Symposium on Software Reliability Engineering (ISSRE), pp 24-29, 2018, **CORE-A, Best Industrial Paper Award**,CCF-B;
- C.5 Zhou Xu, Shuai Li, **Yutian Tang**, Xiapu Luo, Tao Zhang, Jin Liu, Jun Xu, "Cross Version Defect Prediction with Representative Data via Sparse Subset Selection", 26th IEEE/ACM International Conference on Program Comprehension (ICPC), 2018, CORE-A,CCF-B;
- C.4 **Yutian Tang**, Hareton Leung, "Constructing Feature Model by Identifying Variability-aware Modules", 25th IEEE/ACM International Conference on Program Comprehension (ICPC), pp. 263-274, 2017, CORE-A,CCF-B;
- C.3 **Yutian Tang**, Hareton Leung, "StiCProb: A Novel Feature Mining Approach using Conditional Probability", 24th IEEE International Conference on Software Analysis, Evolution, and Reengineering (SANER), pp. 45-55, 2017, **CORE-A,CCF-B**
- C.2 **Yutian Tang**, Hareton Leung, "Feature Mining for Product Line Construction", The First International Conference on Advances and Trends in Software Engineering (SOFTENG), pp.29-33, 2015;
- C.1 **Yutian Tang**, Hareton Leung, "A Top-down Feature Mining Framework for Software Product Line", 17th International Conference on Enterprise Information Systems (ICEIS), pp.71-81, 2015;

Refereed Journal Articles (23)

- J.23 Yongkun Liu, Jiachi Chen, Tingting Bi, John Grundy, Yanlin Wang, Ting Chen, **Yutian Tang**, Zibin Zheng, "An Empirical Study on Low Code Programming using Traditional vs Large Language Model Support", *Accepted to Journal of Systems and Software*, 2025, CCF-B;
- J.22** Zhijie Liu^(stu.), **Yutian Tang**^{*}, Meiyun Li, Xin Jin, Yunfei Long, Liang Feng Zhang, Xiapu Luo "LLM-CompDroid: Repairing Configuration Compatibility Bugs in Android Apps with Pre-trained Large Language Models", *ACM Transactions on Software Engineering and Methodology*, 2025, **Top-1 Journal in SE field, IF=6.6, JCR-Q1, CCF-A**

- J.21 Yutian Tang**, Xiapu Luo, Yuming Zhou, "A Systematic Study on Real-world Android App Bundles", IEEE Transactions on Software Engineering, 2025, **Top-1 Journal in SE field, CCF-A, JCR-Q1**
- J.20 Xutong Liu, Yufei Zhou, **Yutian Tang**, Junyan Qian, Yuming Zhou. "Human-in-the-loop online just-in-time software defect prediction: What have we achieved and what do we still miss?", Science of Computer Programming, CCF-B, 2025
- J.19** Jianzhong Su, Jiachi Chen, Zhiyuan Fang, Xingwei Lin, **Yutian Tang**, and Zibin Zheng. "SmartOracle: Generating Smart Contract Oracle via Fine-Grained Invariant Detection." IEEE Transactions on Software Engineering, 2025, **Top-1 Journal in SE field, JCR-Q1, CCF-A**
- J.18** Jinan Jiang, Zihao Li, Haoran Qin, Muhui Jiang, Xiapu Luo, Xiaoming Wu, Haoyu Wang, **Yutian Tang**, Chenxiong Qian, Ting Chen, "Unearthing Gas-Wasting Code Smells in Smart Contracts with Large Language Models", IEEE Transactions on Software Engineering, 2024, **Top-1 Journal in SE field, CCF-A, JCR-Q1**
- J.17** Zhichao Zhou^(stu.), Yuming Zhou, Chunrong Fang, Zhenyu Chen, Xiapu Luo, Jingzhu He, **Yutian Tang***, "Coverage Goal Selector for Combining Multiple Criteria in Search-Based Unit Test Generation", *IEEE Transactions on Software Engineering*, 2024, **Top-1 Journal in SE field, IF=9.322, JCR-Q1, CCF-A**
- J.16** Zhijie Liu^(stu.), **Yutian Tang***, Xiapu Luo, Yuming Zhou, Liang Feng Zhang, "No Need to Lift a Finger Anymore? Assessing the Quality of Code Generation by ChatGPT", *IEEE Transactions on Software Engineering* **Top-1 Journal in SE field, IF=9.322, JCR-Q1, CCF-A**
- J.15** **Yutian Tang***, Zhijie Liu^(stu.), Zhichao Zhou^(stu.), Xiapu Luo, "ChatGPT vs SBST: A Comparative Assessment of Unit Test Suite Generation", *IEEE Transactions on Software Engineering*, 2024, **Top-1 Journal in SE field, IF=9.322, JCR-Q1, CCF-A**
- J.14** Yang Wang, Peng Zhang, Zeyu Lu, Yibiao Yang, **Yutian Tang**, and Yuming Zhou, "Uncovering bugs in code coverage profilers via control flow constraint solving", *IEEE Transactions on Software Engineering*, 2023, **Top-1 Journal in SE field, IF=9.322, JCR-Q1, CCF-A**
- J.13 Yuanqing Mei, Yi Rong, Shiran Liu, Zhaoqiang Guo, Yibiao Yang, Hongmin Lu, **Yutian Tang**, and Yuming Zhou, "Deriving thresholds of object-oriented metrics to predict defect-proneness of classes: A large-scale meta-analysis", *International Journal of Software Engineering and Knowledge Engineering*, 33, no. 05 (2023): 651-695 IF=1.01, CCF-C
- J.12** Hongchen Chao^(stu.), Shuai Li, Yuming Zhou, Ming Fan, Xuejiao Zhao, **Yutian Tang***, "Cheating Your Apps: Black-box Adversarial Attacks on Deep Learning Apps", *Journal of Software: Evolution and Process*, 2023, CCF-B, IF=1.932
- J.11** Le Yu, Haoyu Wang, Xiapu Luo, Tao Zhang, Kang Liu, Jiachi Chen, Hao Zhou, **Yutian Tang**, Xusheng Xiao, "Towards Automatically Localizing Function Errors in Mobile Apps with User Reviews", *IEEE Transactions on Software Engineering*, 49, no. 4 (2022): 1464-1486 **Top-1 Journal in SE field, CCF-A, IF=9.322, JCR-Q1**

- J.10 Jian Huang, Shanhui Liu, **Yutian Tang**^{*}, Xiushan Zhang, "Object-Level Remote Sensing Image Augmentation Using U-Net-Based Generative Adversarial Networks", *Wireless Communication and Mobile Computing*, 2021 (2021): 1-12, CCF-C, IF=2.146
- J.9 Yutian Tang**, Hao Zhou, Xiapu Luo, Ting Chen, Haoyu Wang, Zhou Xu, and Yan Cai, "XDebloat: Towards Automated Feature-Oriented App Debloating", *IEEE Transactions on Software Engineering*, 48, no. 11 (2021): 4501-4520 **Top-1 Journal in SE field, IF=9.322, JCR-Q1, CCF-A**
Top-1 Recommended Paper from Bing AI for app debloating topic
- J.8 Zhen Yang, Jacky Keung, Md Alamgir Kabir, Xiao Yu, **Yutian Tang**, Miao Zhang, Shuo Feng, "AComNN: Attention Enhanced Compound Neural Network For Financial Time-Series Forecasting With Cross-Regional Features", *Applied Soft Computing*, 111 (2021): 107649. **JCR-Q1, IF=5.47**
- J.7 Kunsong Zhao, Zhou Xu, Tao Zhang, **Yutian Tang**, Meng Yan, "Simplified Deep Forest Model based Just-In-Time Defect Prediction for Android Mobile Apps", *IEEE Transactions on Reliability*, 70, no. 2 (2021): 848-859, **JCR-Q1, CCF-B, IF=5.883**
- J.6 Yutian Tang**^{*}, Haoyu Wang, Xian Zhan, Xiapu Luo^{*}, Yajin Zhou, Hao Zhou, Qiben Yan, Yulei Sui, Jacky Keung, "A Systematical Study on Application Performance Management Libraries for Apps", *IEEE Transactions on Software Engineering*, 2021, 48(8), pp. 3044 - 3065 **Top-1 Journal in SE field, CCF-A, JCR-Q1, IF=9.322;**
- J.5 Zhou Xu, Tao Zhang, Jacky Keung, Meng Yan, Xiapu Luo, Xiaohong Zhang, Ling Xu, **Yutian Tang**, "Feature Selection and Embedding Based Cross Project Framework for Identifying Crashing Fault Residence" *Information and Software Technology (IST)*, 131 (2021): 106452, **JCR-Q1, CCF-B, IF=3.862**
- J.4 Zhou Xu, Shuai Li, Jun Xu, Jin Liu, Xiapu Luo, Yifeng Zhang, Tao Zhang, Jacky Keung, **Yutian Tang**, "LDFR: Learning deep feature representation for software defect prediction", *Journal of Systems and Software (JSS)*, 158 (2019), 110402 **JCR-Q1, CCF-B, IF=3.514**
- J.3 Zhou Xu, Shuai Pang, Tao Zhang, Xiapu Luo, **Yutian Tang**, Xiao Yu, Lei Xue, "Cross Project Defect Prediction via Balanced Distribution Adaptation based Transfer Learning", *Journal of Computer Science and Technology (JCST)*, 34 (2019): pp. 1039-1062, IF=1.87, CCF-B
- J.2 Zhou Xu, Shuai Li, Xiapu Luo, Jin Liu, Tao Zhang, **Yutian Tang**, Jun Xu, Peipei Yuan, Jacky Keung, "TSTSS: A Two-Stage Training Subset Selection Framework for Cross Version Defect Prediction", *Journal of System and Software (JSS)*, 154 (2019): pp. 59-78, **JCR-Q1; IF=3.514, CCF-B**
- J.1 Zhou Xu, Jin Liu, Xiapu Luo, Zijiang Yang, Yifeng Zhang, Peipei Yuan, **Yutian Tang**, Tao Zhang, "Software Defect Prediction Based on Kernel PCA and Weighted Extreme Learning Machine", *Information and Software Technology (IST)*, 106 (2019), pp. 182-200. **JCR-Q1; IF=3.862, CCF-B**

Software Copyright

- C.1 **Yutian Tang**, Zhijie Liu, “Big Data-Based Mobile Application Malicious Component Detection System V1.0”, Software Copyright, 2023, Authorized by: National Copyright Administration of P.R.China.

Patents

- P.1 **Yutian Tang**, Zhijie Liu, Liangfeng Zhang, “Advanced Methods for Detecting Malicious Elements in Mobile Apps Using Big Data Analytics”, Chinese Patent.

Software

- S.1 **Yutian Tang**, WorkBuddy, Google Chrome Extension, 2024, <https://chromewebstore.google.com/detail/workbuddy/hihenendbjafpikbhakolegagkamdgjo>
- S.2 **Yutian Tang**, Healthy Eye Reminder, Google Chrome Extension, 2024, <https://chromewebstore.google.com/detail/healthy-eye-reminder/kmjheafpaclgijopjfoofbaodmiglkhd?hl=en&authuser=0>

Impact and Engagement

International Stands Working Group/Committee

Standard Working Group Member **IEEE Standard P1633 Working Group: (Recommended Practice for Software Reliability)**

Grant/Award Review Panelist

Fellowship Accessor **Green Future Fellowships Assessor, 2024, UK**

Grant Reviewer **Engineering and Physical Sciences Research Council (EPSRC) Standard Research Grant 2024, UK**

Grant Reviewer **Engineering and Physical Sciences Research Council (EPSRC) New Investigator Award 2024, UK**

Member **Engineering and Physical Sciences Research Council (EPSRC) Peer Review College (Members will service as grant review panelists for EPSRC proposals), UK EPSRC Research Grant Review Panelist 2023 - now**

Award Panelist **Shanghai Science and Technology Award (Top-1 award in Shanghai in field of Sci. and Tech.), 2022**

Grant Review Panelist **Shanghai Innovation Grant for Small and medium-sized Enterprises, 2021**

Grant Review Panelist **National Natural Science Foundation of China (NSFC), 2023 - now**

Grant Review Panelist **National Natural Science Foundation of China (NSFC), 2025: General Program**

Interview/Assessment Panel

SyMeCo Lero, the SFI Research Centre for Software, 2025
 Marie Skłodowska-Curie Postdoctoral Fellowship Programme

Review Panel IEEE Senior Member Virtual Review Panel 2025

Editorship

Review Editor **Frontiers in Computer Science** (ESCI Indexed), 2021-now

Review Editor **Scientific Programming**, 2022-now

Long-term School and University Admin Roles

Co-Impact Education and Practice (EAP) Section, School of Computing Science, University of
 Champion Glasgow, 10/2024 - now

Career Liaison School of Computing Science, University of Glasgow, 10/2025 - now
 Officer

Technical Committee Member & Interest Group

Committee **CCF-Technical Committee of Software Engineering**
 Member

Committee **CCF-Technical Committee of System Software**
 Member

Committee **Shanghai Computer Society-Technical Committee of Software Engineering**
 Member

Technical **IEEE Technical Committee on Software Engineering (TCSE)**
 Committee
 Member

Trustworthy **The Alan Turing Institute**
 Digital
 Identity
 Interest
 Group

Interpretation, **The Alan Turing Institute**
 Verification
 and
 Modelling
 Interest
 Group

Organizing Committee Member

2025 **[1] International Conference on Evaluation and Assessment in Software En-
 gineering (EASE) 2026**, Doctoral Consortium Co-Chair

2024 **[1] 2024**, Publicity Co-Chair

Conference Technical Program Committee (TPC) Member (46 in total)

- 2027 **[1] 49th International Conference on Software Engineering (ICSE) Core A*, Top-1 in Software Engineering**
Research Track
- 2026 **[1] 48th International Conference on Software Engineering (ICSE) Core A*, Top-1 in Software Engineering**
Research Track
[2] ACM International Conference on the Foundations of Software Engineering (FSE) (formerly known as ESEC/FSE) **Core A***
Research Track
[3] Mining Software Repositories 2026 Core A
Research Track
[4] International Conference on Program Comprehension (ICPC) 2026 Core A
Research Track
[5] The Web Conference (a.k.a WWW) Core A*
Research Track
[6] IEEE/ACM International Conference on Automated Software Engineering (ASE) Core A*
[7] IEEE Computer Society's flagship conference on Computers, Software, and Applications (COMPSAC) Research Track
- 2025 **[1] 47th International Conference on Software Engineering (ICSE) Core A*, Top-1 in Software Engineering**
Research Track
[2] The IEEE/ACM Automated Software Engineering (ASE) Core A*, Top-1 in Software Engineering
Research Track
[3] Mining Software Repositories 2025 Core A
Research Track
[4] The Web Conference (a.k.a WWW) Core A*
Research Track
[5] IEEE/ACM International Conference on Automated Software Engineering Core A*
[6] 32nd Asia-Pacific Software Engineering Conference (APSEC)
Research Track;
[7] Program Committee, International Conference on Software Engineering & Knowledge Engineering 2024 Core B
Research Track;

- 2024 [1] **ACM International Conference on the Foundations of Software Engineering (FSE) (formerly known as ESEC/FSE)** **Core A***
Research Track
- [2] **39th IEEE/ACM International Conference on Automated Software Engineering (ASE)** **Core A***
Research Track
- [3] **International Working Conference on Variability Modelling of Software-Intensive Systems**
Research Track;
- [4] **International Conference on Program Comprehension (ICPC)** **Core A**
Research Track;
- [5] **The IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)** **Core A**
ERA Track;
- [6] **Program Committee, 36th International Conference on Advanced Information Systems Engineering (CAiSE)** **Core A**
- [7] **Program Committee, ACM TheWebConf 2024** **Core A***
Research Track (System Track);
- [8] **Program Committee, International Conference on Software Engineering & Knowledge Engineering 2024** **Core B**
Research Track;
- [9] **31st Asia-Pacific Software Engineering Conference (APSEC)**
Research Track;

- 2023 [1] **The ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)** **Core A***
 Research Track
 Student Research Competition Track
- [2] **38th IEEE/ACM International Conference on Automated Software Engineering (ASE)** **Core A***
 Research Track
 The New Ideas and Emerging Results (NIER) Track
- [3] **IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)** **Core A**
 Research Track, Tool Demo Track
- [4] **International Conference on Program Comprehension (ICPC)** **Core A**
 Research Track;
- [5] **International Conference on Software Engineering & Knowledge Engineering, (SEKE)** **Core B**
 Research Track;
- [6] **2023 International Conference on Evaluation and Assessment in Software Engineering (EASE)** **Core A**
 Research Track;
- [7] **20th International Conference on Mining Software Repositories (MSR)** **Core A**
 Research Track, Industry Track;
- [8] **Programming Language Design and Implementation (PLDI)** **Core A***
 Artefact Evaluation Track
- [9] **35th International Conference on Advanced Information Systems Engineering (CAiSE) 2023;** **Core A**
 Research Track
- [10] **Program Committee, IEEE/ACM International Conference on Mobile Software Engineering and Systems (MOBILESoft): 2023;**
 Research Track;
- [11] **International Working Conference on Variability Modelling of Software-Intensive Systems**
 Research Track;
- [12] **30th Asia-Pacific Software Engineering Conference (APSEC)**
 Research Track;

- 2022 [1] **The ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)** **Core A***
Artefact Evaluation Track
- [2] **Programming Language Design and Implementation (PLDI)** **Core A***
Artefact Evaluation Track
- [3] **International Conference on Software Engineering (ICSE)** **Core A***
Artefact Evaluation Track
- [4] **International Conference on Program Comprehension (ICPC)** **Core A**
Research Track;
- [5] **IEEE/ACM International Conference on Mobile Software Engineering and Systems (MobileSoft)**
Research Track, Industry Track;
- [6] **International Conference on Software Engineering & Knowledge Engineering, (SEKE)** **Core A**
Research Track;
- [7] **International Conference of Modern Systems Engineering Solutions**
Research Track
- [8] **Future Technologies Conference**
Research Track
- [9] **Computing Conference**
Research Track
- 2021 [1] **The ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)** **Core A***
Artefact Evaluation Track
- [2] **CCF Chinasoft Conference (ChinaSoft)**
Software & Data Intelligence Track
- Journal Reviewer (20 in total)**
- Reviewer **IEEE Transactions on Dependable and Secure Computing, 2023 - now**
- Reviewer **Journal of Computer Security, 2022 - now**
- Reviewer **IEEE Transactions on Reliability, 2022 - now**
- Reviewer **ACM Transactions on Software Engineering and Methodology (TOSEM), 2022 - now**
- Reviewer **IEEE Communication Magazine, 2021;**
- Reviewer **IEEE Transactions on Software Engineering, 2020 - now;**
- Reviewer **Automated Software Engineering (ASE), 2022 - now;**
- Reviewer **Empirical Software Engineering (EMSE), 2022 - now**
- Reviewer **Frontier of Computer Science, 2020 - now**
- Reviewer **SN Applied Sciences, 2022**
- Reviewer **Scientific Programming, 2022 - now**
- Reviewer **IEEE Transactions on Information Forensics and Security, 2023 - now**
- Reviewer **The Journal of System and Software, 2020 - now**
- Reviewer **Frontiers of Information Technology & Electronic Engineering, 2022 - now**
- Reviewer **SCIENCE CHINA Information Sciences, 2021 - now**

Reviewer **IEEE Access, 2017 - now**
Reviewer **IEEE Software, 2024 - now**
Reviewer **Security and Communication Networks, 2023 - now**
Ad-hoc Reviewer **Journal of Information Technology Research, 2020 - now;**
Sub-reviewer **IEEE Transactions on Services Computing, IEEE Transactions on Biometrics, Behavior and Identity Science**

Conference Session Chairs

Session Chair **The ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA), 2025**
Session Chair **IEEE/ACM International Conference on Software Engineering (ICSE), 2025**
Session Chair **ACM International Conference on the Foundations of Software Engineering (FSE), 2024**

Awards & Recognition

2026 **ACM SIGSOFT Distinguished Paper Award @ ICSE**
2025 **Research Software Engineering (RSE) Rising Star Nomination, RSE-UK**
2025 **OpenAI Cybersecurity Grant Award, OpenAI**
2025 **Certificated Reviewer of ACM, ACM**
2025 **Fellow, Recognizing Excellence in Teaching, University of Glasgow, 2025**
2024 **Google Cloud Credit Award, 2024**
2024 **OpenAI Researcher Access Program, 2024**
2023 **Best Reviewer Award, Asia-Pacific Software Engineering Conference (APSEC), 2023**
2021 **Shanghai Pujiang Talent Program, Science and Technology Commission of Shanghai Municipality**
2018 **Best Industry Paper Award International Symposium on Software Reliability Engineering, Paper: "A Smart Context-aware Program Assistant based on Dynamic Programming Event Modeling"**
2013–2017 **Ph.D. Scholarships, The Hong Kong Polytechnic University**
2012 **Undergraduate Research Award, First Prize, Jilin University**
2012 **Undergraduate Scholarships, Second Prize, Jilin University**
2011 **Undergraduate Scholarships, Third Prize, Jilin University**

News Interview/Media Report

VOC, CCTV 12/Aug/2021 – The Voice of China, China Central Television; [see <https://mp.weixin.qq.com/s/glvaDNdsAAXrdAn9imDgnQ> (in Chinese)]

- VOC, CCTV 25/May/2022 – The Voice of China, China Central Television; [see https://content-static.cctvnews.cctv.com/snow-book/index.html?toc_style_id=feeds_default&share_to=wechat&item_id=631265256291590857&track_id=ABAA3E71-3BC8-468E-BB30-E7749E30EBB7_675580162403] (in Chinese)]
- VOC, CCTV 26/June/2022 – The Voice of China, China Central Television; [see https://content-static.cctvnews.cctv.com/snow-book/index.html?toc_style_id=feeds_default&share_to=wechat&item_id=7098209212896444086&track_id=9CEB9984-7D6A-4A10-8AF3-73D7EADCA068_677671690278] (in Chinese)]
- IEEE Spectrum 6/July/2024 – IEEE Spectrum [see <https://spectrum.ieee.org/chatgpt-for-coding>]

Professional Qualifications & Membership

- 2023–Present **RISCS Affiliate Fellowship**, The Research Institute for Sociotechnical Cyber Security (RISCS), (funded by the National Cyber Security Centre (NCSC))
- 2023–Present **Senior Member**, IEEE
- 2022–Present **Member**, ACM
- 2022–Present **Member**, European Alliance for Innovation (EAI)
- 2020–Present **Member**, China Computer Federation (CCF)
- 2020–Present **Member**, EuroSys
- 2017–Present **Member**, ISACA
- 2012–Present **Senior Software Engineer**, under “China Qualification Certificate of Computer Science Technology Proficiency”

Teaching

Instructor

- Uni. of Glasgow COMPSCI5093/5104 Secured Software Engineering, University of Glasgow; Spring'24 (130+ students), Spring'25 (200+ students), Spring'26 (250+ students)
- Uni. of Glasgow COMPSCI4015 Professional Software Development, University of Glasgow; Fall'23, Spring'24, Fall'24, Spring'25, Fall'25, Spring'26 (400+ students)
- Uni. of Glasgow COMPSCI5074 MSc IT + Masters Team Project, University of Glasgow; Spring'26 (170+ students)

ShanghaiTech **CS132**, Software Engineering, ShanghaiTech University.

Uni.	Year (Sem)	Enrollment	Teaching Evaluation
	2021 (Spring)	129	4.5/5.0
	2021 (Fall)	29	4.7/5.0

ShanghaiTech **CS224**, Program Analysis, ShanghaiTech University.

Uni.	Year (Sem)	Enrollment	Teaching Evaluation
	2022 (Spring)	26	4.9/5.0

Teaching Assistant

- PolyU.HK COMP3235, Software Project Management, Department of Computing; (Course for both Undergraduate);
- PolyU.HK COMP3233, Software Testing and Quality Assurance, Department of Computing; (Course for both Undergraduate);
- PolyU.HK COMP4911, Capstone Project, Department of Computing; (Course for both Undergraduate);
- PolyU.HK COMP3211, Software Engineering, Department of Computing; (Course for both Undergraduate);
- PolyU.HK COMP309, System Programming, Department of Computing; (Course for both Undergraduate);
- PolyU.HK COMP322, Enterprise Information Systems Project Implementation, Department of Computing (Course for both Undergraduate);

Supervision

Summary I have supervised 9 research-oriented students, 14 graduate apprenticeship students, and 11 project/thesis-oriented students.

Ph.D. Students and MPhil Students (By Research) (8 Students)

- 2024-2028 Yuxi Chen (University of Glasgow)-Main Supervisor
- 2026-2029 Paul Smith (University of Glasgow)-Main Supervisor
- 2024-2028 Susmita Das (University of Glasgow)-Second Supervisor
- 2024-2026 Bernie Chen (University of Glasgow)-Second Supervisor (with UofG Singapore)
- 2023 Jinan Jiang (with H.K Polytechnic University, Co-supervised with Prof. Xiapu Luo) TSE24
- 2022-2025 Zhijie Liu (with ShanghaiTech Uni.)
Publication: ASE23,ISSTA25,TOSEM25,TSE24
- 2022-2025 Haitao Hu (with ShanghaiTech Uni.)
- 2021-2024 Zhichao Zhou (with ShanghaiTech Uni.)
Publication: ASE22,OOPSLA25,TSE24a,TSE24b

Visiting PhD Students

- 2025.04-06 David Romero Organvdez, Ph.D student from University of Seville (Universidad de Sevilla), Spain
Publication: [C24]

Research Intern (2 Students)

- 2020-2022 Hongchen Cao, Undergraduate from ShanghaiTech Uni.
Publication: [J12][C22]
- 2022-2023 Xingchen Jin, Undergraduate from ShanghaiTech Uni.
Visiting Student at University of California, Berkeley (UCB)

Graduate Apprenticeships Workplace Assessment Year 3 (7 Students in total)

- 2024-2025 3 Year-3 Students, University of Glasgow
- 2023-2024 4 Year-3 Students, University of Glasgow

Graduate Apprenticeships Workplace Assessment Year 4 (7 Students in total)

- 2024-2025 3 Year-4 Students, University of Glasgow
- 2023-2024 4 Year-4 Students, University of Glasgow

Undergraduates/Msc/Projects/Thesis (11 Students in total)

- 2023-2024 2 Year-4 Students, 4 Msc Students, University of Glasgow
- 2022 1 Final Year Student, ShanghaiTech University
- 2021 3 Final Year Students, ShanghaiTech University
- 2020 1 Final Year Student, ShanghaiTech University

Selected University/Teaching/ECDP Trainings

- 2023-2025 Postgraduate Certificate in Academic Practice (PgCAP)
- 2023 Equality and Diversity Essentials, University of Glasgow; 07/2023
- 2023 Implementing Reasonable Adjustments, University of Glasgow; 07/2023
- 2023 Information Security, University of Glasgow; 07/2023
- 2023 Introduction to the General Data Protection Regulation, University of Glasgow; 07/2023

University & Department Services

- 2023 University Open Day, School of Computing Science, University of Glasgow; 30/08/2023
- 2023 Application Evaluation for Review, School of Computing Science, University of Glasgow; 10/08/2023
- 2021 Undergraduate Admission Coordinator, ShanghaiTech University, Hangzhou (Zhejiang); 28/6/2021-1/7/2021
- 2021 Graduate Recruiting at Jilin University, Changchun, 11/6/2021 (Online)
- 2021 Undergraduate Admission Coordinator, ShanghaiTech University, Tianjin City; 24/4/2021, 25/4/2021
- 2021 PI Interview Committee, System and Security Center, 1C-201, SIST, ShanghaiTech University, 4/3/2021; 11/3/2021
- 2021 Graduate Admission Interview, School of Information Science and Technology, 1A-302, SIST, ShanghaiTech University, 19/3/2021
- 2020 Modular High Performance Computing Application Innovation Workshop, 26/11/2020-Room 110, Administration Center, ShanghaiTech University
- 2020 Graduate Admission Interview, School of Information Science and Technology, 1A-212, SIST, ShanghaiTech University, 23/10/2020;13/09/2020
- 2020 Undergraduate Seminar, "System Security, Are We There Yet?", School of Information Science and Technology, ShanghaiTech University, 6/11/2020
- 2020 PI Interview Committee, System and Security Center, 1C-201, SIST, ShanghaiTech University, 19/11/2020

Conference Talks & Invited Talks

- UCL CREST Workshop **Yutian Tang**, Invited Speaker at UCL CREST workshop, 2025, Workshop;
- Fudan University **Yutian Tang**, "Demystifying Application Performance Management Libraries for Android", **Yutian Tang**, Fudan University, 8/4/2021, Seminar;
- ShanghaiTech University **Yutian Tang**, "System Security, Are We There Yet?", **Yutian Tang**, School of Information Science and Technology, ShanghaiTech University, 6/11/2020, Undergraduate Seminar;
- Invited Speaker The 12th International Conference on Advanced Infocomm Technology(2020);
- ShanghaiTech University **Yutian Tang**, "Android, Don't Lie to Me", **Yutian Tang**, ShanghaiTech University, China;
- Queen's University Belfast **Yutian Tang**, "Android, Don't Lie to Me", **Yutian Tang**, Queen's University Belfast, United Kingdom;
- University of East London **Yutian Tang**, "Android, Don't Lie to Me", **Yutian Tang**, University of East London, United Kingdom;
- University of Dundee **Yutian Tang**, "Android, Don't Lie to Me", **Yutian Tang**, University of Dundee, United Kingdom;
- University of Central Lancashire **Yutian Tang**, "Android, Don't Lie to Me", **Yutian Tang**, University of Central Lancashire, United Kingdom;
- University of Canterbury **Yutian Tang**, "Android, Don't Lie to Me", **Yutian Tang**, University of Canterbury, New Zealand;
- ESEC/FSE'20 **Yutian Tang**, "All Your App Links are Belong to Us: Understanding the Threats of Instant Apps based Attacks", **Yutian Tang**, in *The ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*;
- Tongji University **Yutian Tang**, "All Your App Links are Belong to Us: Understanding the Threats of Instant Apps based Attacks", **Yutian Tang**, School of Software, Tongji University;
- ASE'19 **Yutian Tang**, "Demystifying Application Performance Management Libraries for Android,", **Yutian Tang** in *34th IEEE/ACM International Conference on Automated Software Engineering*;
- SANER'17 **Yutian Tang**, "StiCProb: A Novel Feature Mining Approach Using Conditional Probability", **Yutian Tang**, in *Proceedings of 24th IEEE International Conference on Software Analysis, Evolution, and Reengineering*;
- ICPC'17 **Yutian Tang**, "Constructing Feature Model by Identifying Variability-aware Modules", **Yutian Tang**, in *IEEE/ACM International Conference on Program Comprehension*;
- SOFTENG'15 **Yutian Tang**, "Feature Mining for Product Line Construction", **Yutian Tang**, in *The First International Conference on Advances and Trends in Software Engineering*;
- ICEIS'15 **Yutian Tang**, "Top-down Feature Mining Framework for Software Product Line," **Yutian Tang**, in *Proceedings of the International Conference on Enterprise Information System*;

Languages

Chinese	Native
English	Proficient (CEFR C1)
Cantonese	Basic

References

Available upon requests.